

Protocol Analysis Report

Dataset: Файл данных fd442...

Protocol: Первичка 26.02.26 clean simple protocol (Holm)

Dataset ID: fd4423cd-f069-4e0a-86f7-ae26428741f6

Reproducibility and provenance

Run ID: run_20260302_145605_433411_8d38e44e

Source file: -

Source path: workspace/datasets/fd4423cd-f069-4e0a-86f7-ae26428741f6/source/original.raw

Import settings: sheet=-; header_row=-

Analysis dataset: rows=4, cols=13, xlsx=analysis_dataset.xlsx, parquet=analysis_dataset.parquet

Frozen analysis set: id=-; n_selected=-; mode=-; enforce=-

Reproducibility: ready=True; script=reproduce_run.py; payload=reproduce_payload.json;

manifest=reproducibility_manifest.json

Bootstrap trace artifact: -

Hypothesis discovery artifact: hypothesis_discovery.json

Verification status: passed

Steps excluded from report: 0

Protocol validation: status=passed; enabled=yes; strict=no

Protocol validation summary: checked=14; failed=0; warnings=0; global_errors=0

Validation policy: profile=focused; multiplicity=Holm; repair=Holm; reflection=yes

Multiplicity policy: correction=Holm; post_hoc=Holm; applied_steps=0

Bootstrap policy: enabled=no; samples=1000; applied_steps=0

Validation

artifact:

workspace/datasets/fd4423cd-f069-4e0a-86f7-ae26428741f6/analysis/run_20260302_145605_433411_8d38e44e/artifacts/protocol_validation.json

Artifacts

path:

workspace/datasets/fd4423cd-f069-4e0a-86f7-ae26428741f6/analysis/run_20260302_145605_433411_8d38e44e/artifacts

Protocol Validation

Validation status: passed

Policy profile: focused

Validator mode: enabled=yes; strict=no

Correction policy: multiplicity=Holm; repair=Holm; reflection=yes

Multiplicity policy: correction=Holm; post_hoc=Holm; applied_steps=0

Bootstrap policy: enabled=no; samples=1000; applied_steps=0

Step summary: total=14; checked=14; failed=0; warnings=0; global_errors=0

No critical step findings were detected.

Methods

paired wide: n_steps=10; steps=p_d1_no_gel, p_d2_no_gel, p_d1_gel_d1, p_d2_gel_d1, p_d1_gel_d2, p_d2_gel_d2, p_d1_ctrl_day1_vs_day2, p_d1_exp_day1_vs_day2; targets=-

Bland-Altman analysis: n_steps=4; steps=ba_d1_gel_d1, ba_d2_gel_d1, ba_d1_gel_d2, ba_d2_gel_d2; targets=-

Hypothesis discovery and traceability

Mode: focused; Design: repeated_measures_wide; Total: 3; Covered by steps: 0

Supported: 0; Not supported: 0; Not evaluated: 3

- Сравнить replicate между группами d1_gel_day1_3_5ml_control

H0: Распределение/среднее replicate не различается между уровнями d1_gel_day1_3_5ml_control.

H1: replicate различается минимум между двумя уровнями d1_gel_day1_3_5ml_control.

Method: t_test_ind / mann_whitney / anova; steps: -

Verdict: not evaluated; evidence: -

- Проверить связь между replicate и d1_no_gel_4ml_control
H0: Линейной/монотонной связи между replicate и d1_no_gel_4ml_control нет.
H1: Между replicate и d1_no_gel_4ml_control есть статистическая связь.
Method: pearson / spearman; steps: -
Verdict: not evaluated; evidence: -
- Проверить тренд replicate во времени (d1_gel_day1_3_5ml_control)
H0: Во времени (d1_gel_day1_3_5ml_control) для replicate систематического тренда нет.
H1: Для replicate есть значимый временной тренд.
Method: time_series_analysis / mixed_effects; steps: -
Verdict: not evaluated; evidence: -

Results

Step: Donor1 no gel 4ml: control vs experiment

ID: p_d1_no_gel

Type:

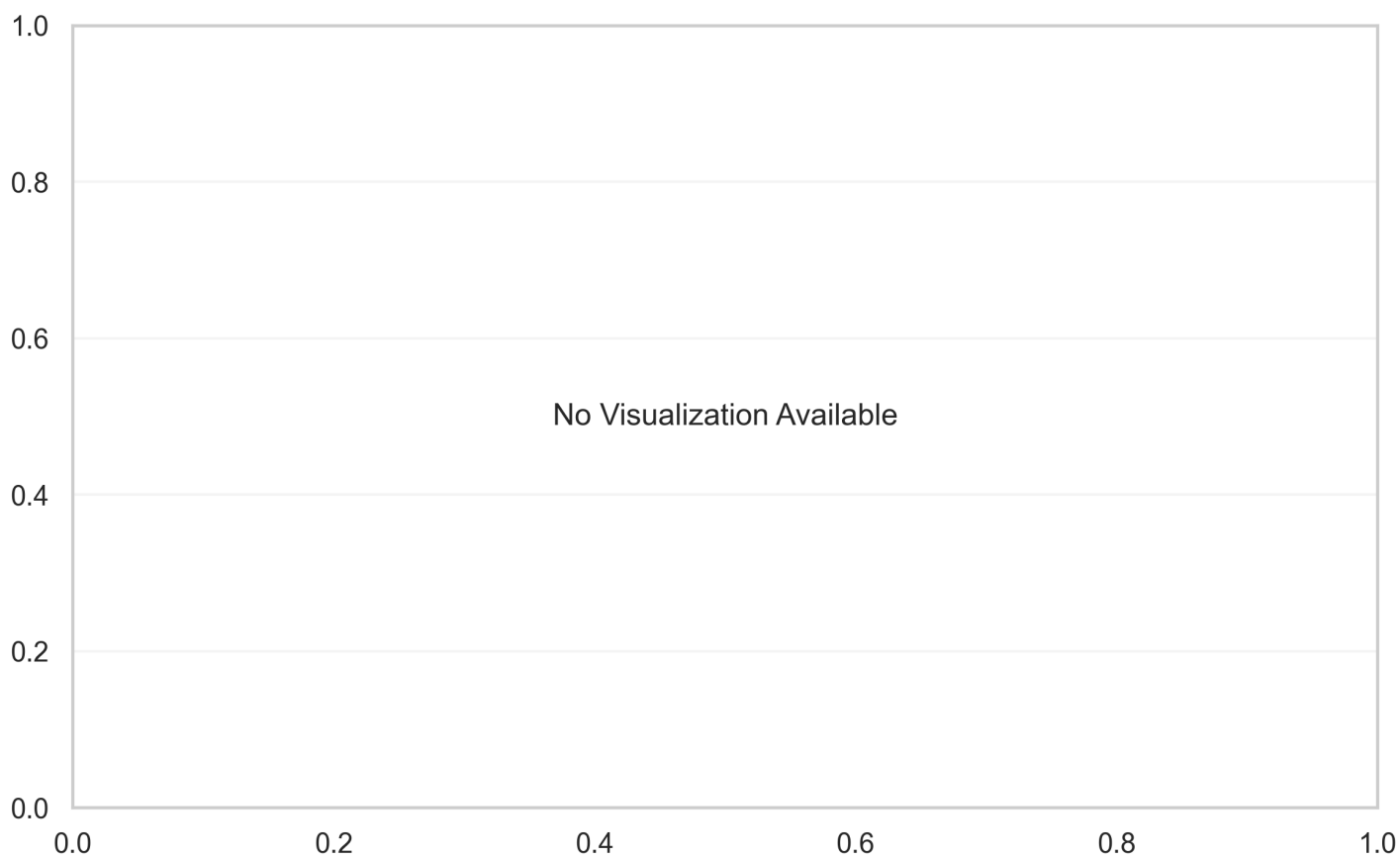
Method: paired wide

p-value: 0.1250

Statistic: 0.000

Effect size: rbc -1.00

Plot: next page



Step: Donor2 no gel 4ml: control vs experiment

ID: p_d2_no_gel

Type:

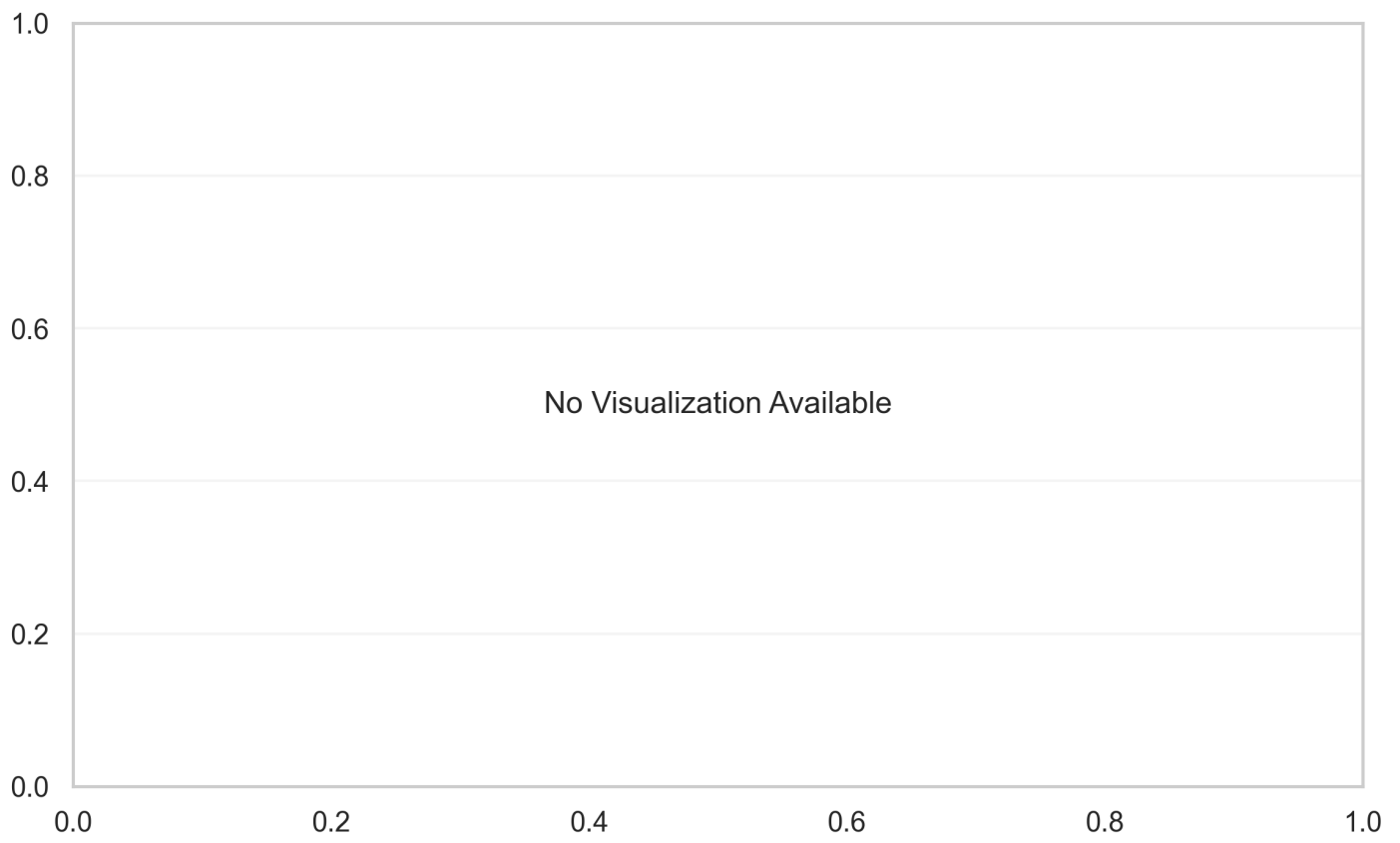
Method: paired wide

p-value: 0.6250

Statistic: 3.000

Effect size: rbc -0.40

Plot: next page



Step: Donor1 gel day1 3.5ml: control vs experiment

ID: p_d1_gel_d1

Type:

Method: paired wide

p-value: 1.0000

Statistic: 5.000

Effect size: rbc 0.00

Plot: next page

Step: Donor2 gel day1 3.5ml: control vs experiment

ID: p_d2_gel_d1

Type:

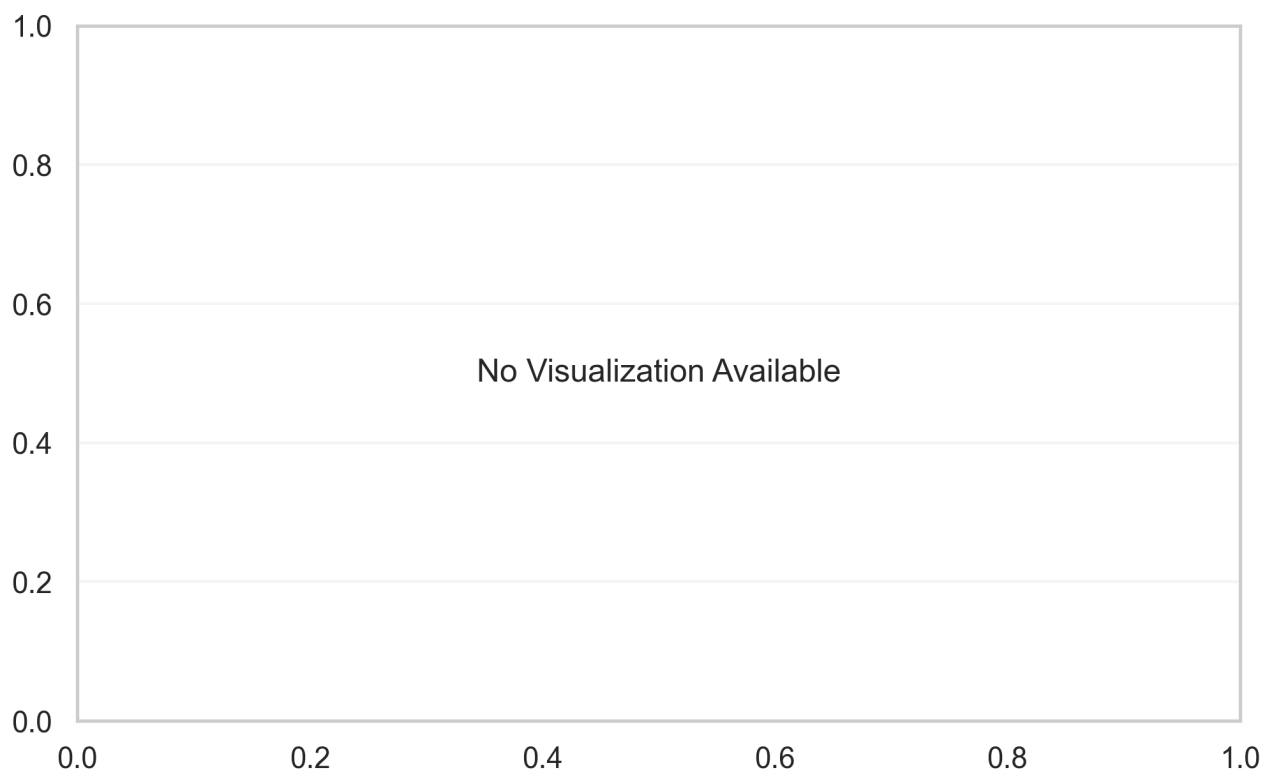
Method: paired wide

p-value: 0.4227

Statistic: 1.000

Effect size: rbc -0.67

Plot: next page



Step: Donor1 gel day2 3.5ml: control vs experiment

ID: p_d1_gel_d2

Type:

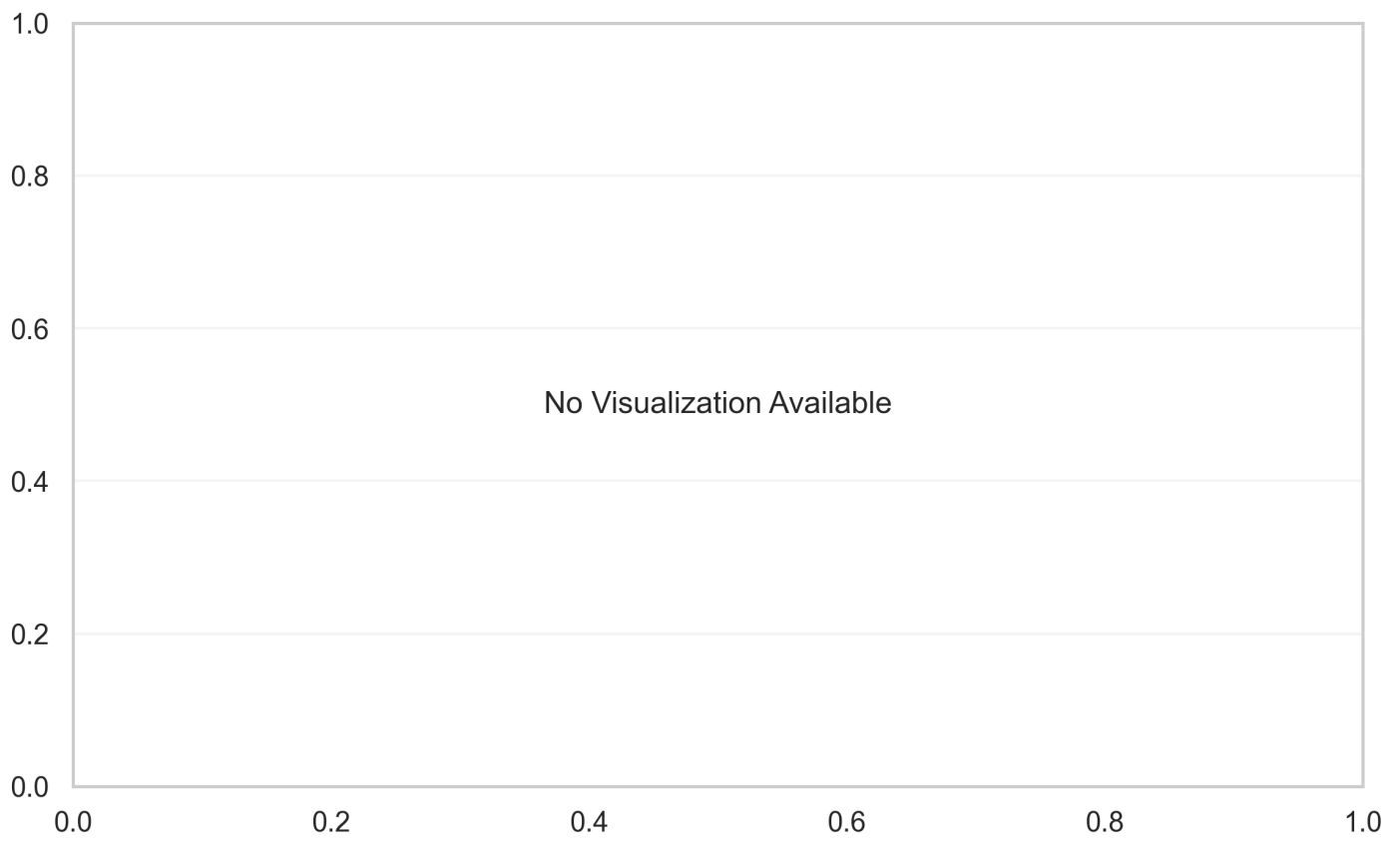
Method: paired wide

p-value: 0.1814

Statistic: 0.000

Effect size: rbc -1.00

Plot: next page



Step: Donor2 gel day2 3.5ml: control vs experiment

ID: p_d2_gel_d2

Type:

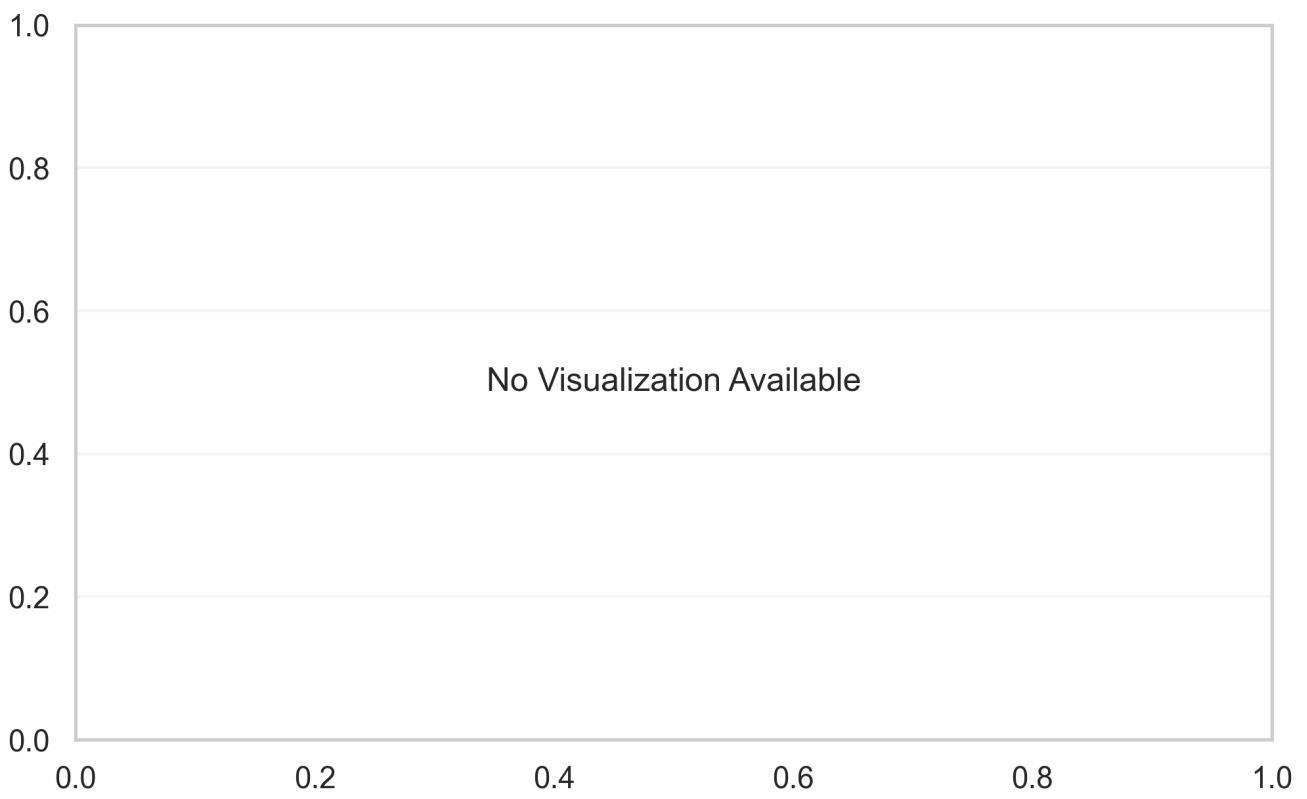
Method: paired wide

p-value: 0.1736

Statistic: 0.000

Effect size: rbc -1.00

Plot: next page



Step: Donor1 control gel: day1 vs day2

ID: p_d1_ctrl_day1_vs_day2

Type:

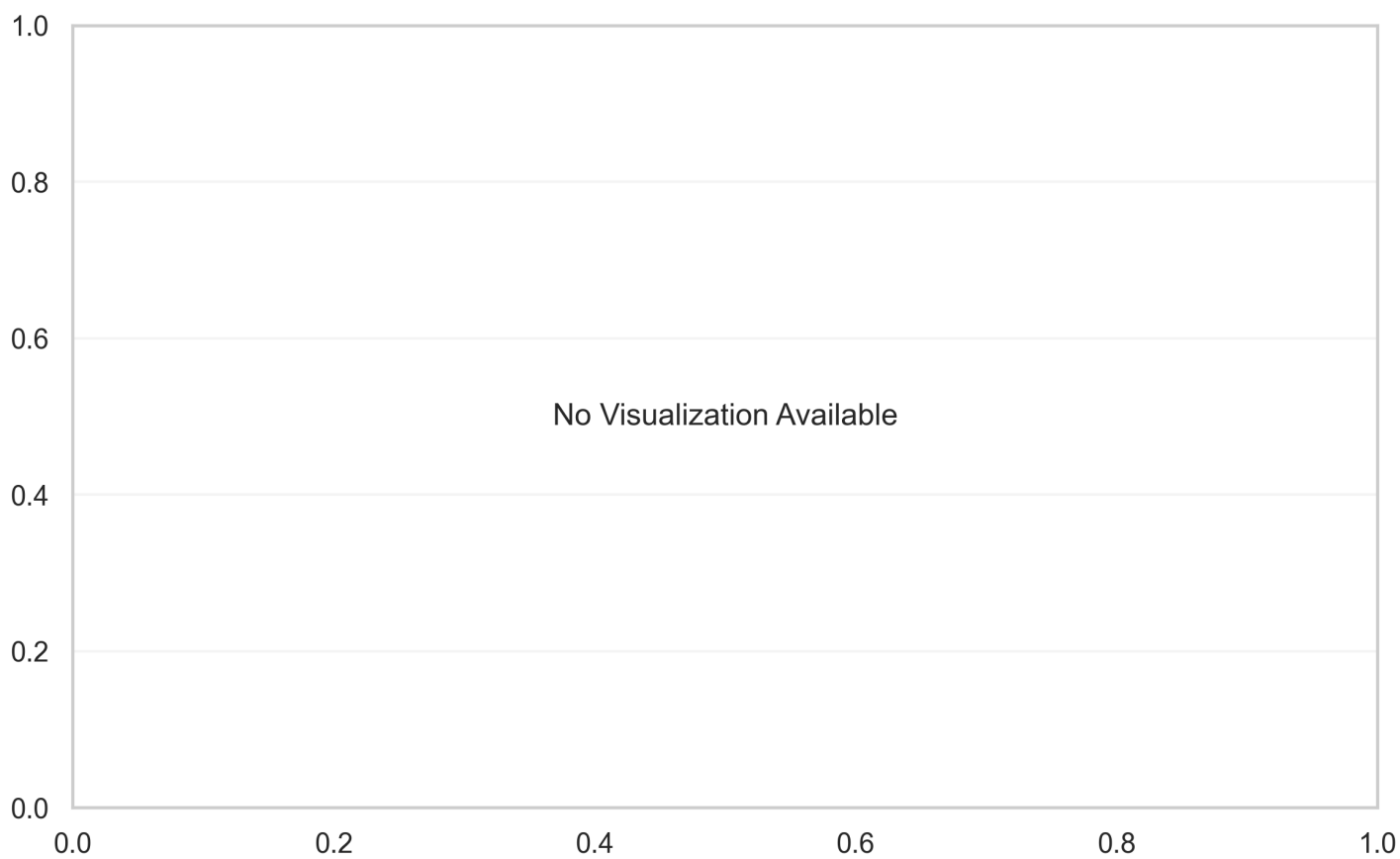
Method: paired wide

p-value: 1.0000

Statistic: 5.000

Effect size: rbc 0.00

Plot: next page



Step: Donor1 experiment gel: day1 vs day2

ID: p_d1_exp_day1_vs_day2

Type:

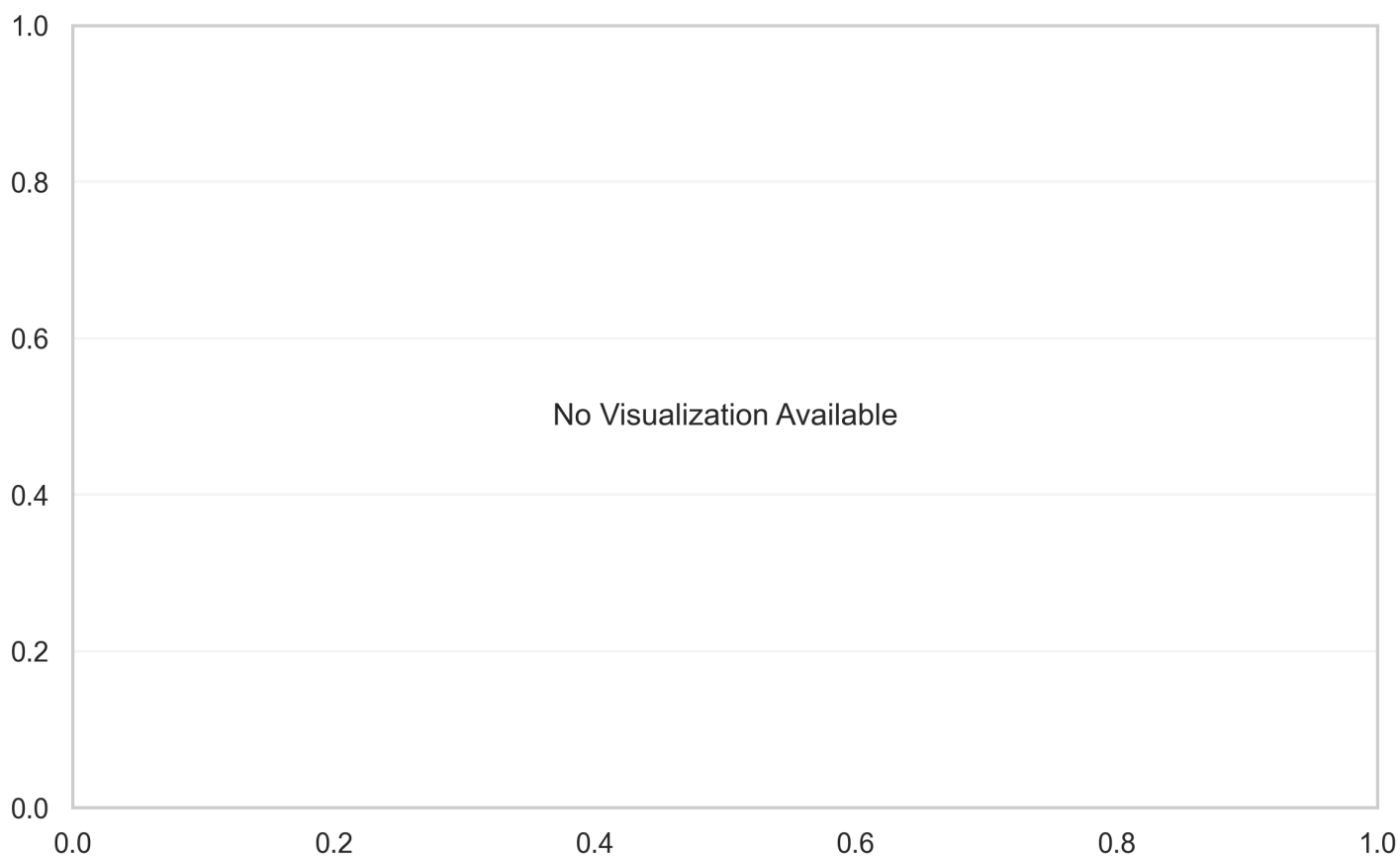
Method: paired wide

p-value: 0.1250

Statistic: 0.000

Effect size: rbc -1.00

Plot: next page



Step: Donor2 control gel: day1 vs day2

ID: p_d2_ctrl_day1_vs_day2

Type:

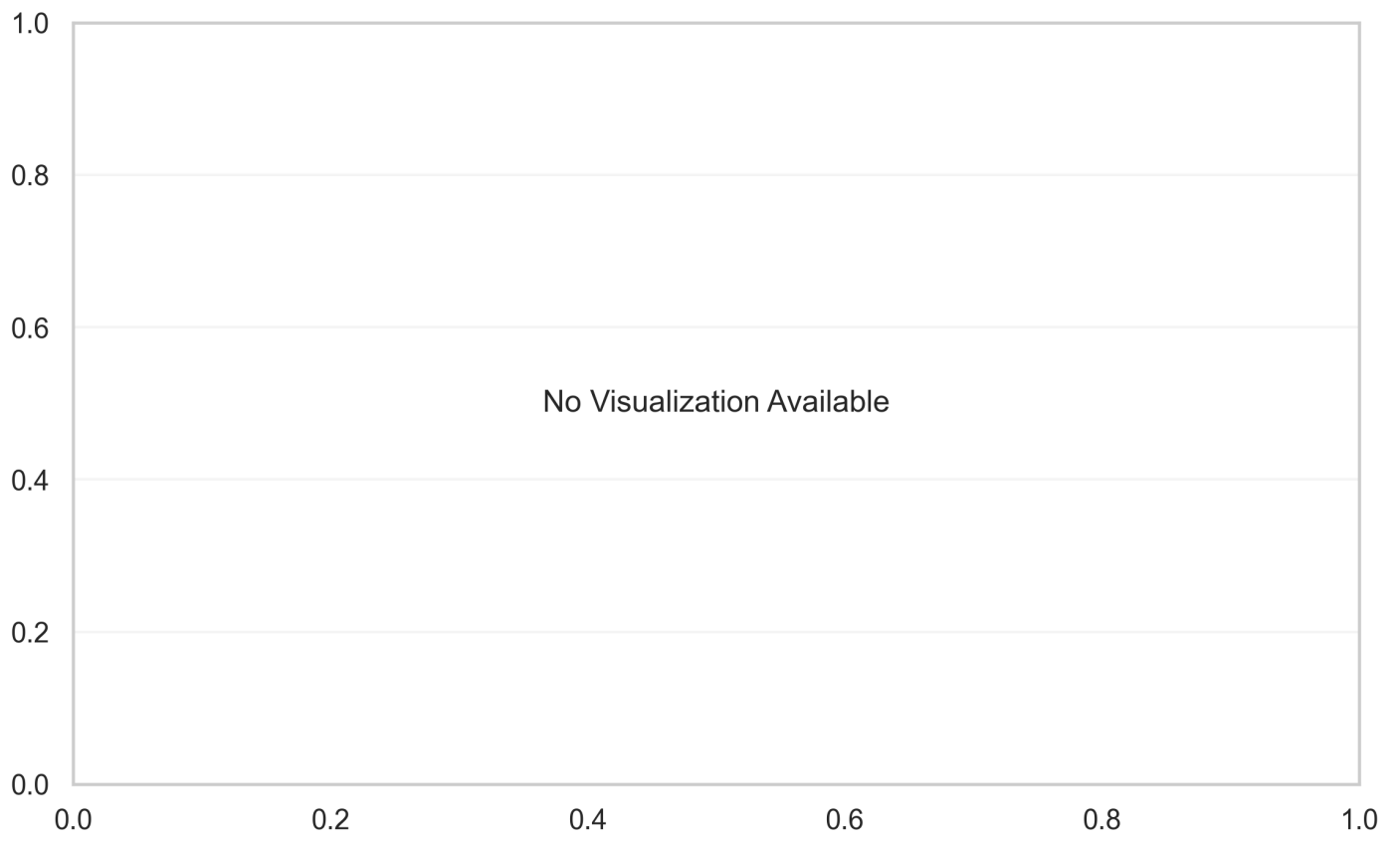
Method: paired wide

p-value: 0.1250

Statistic: 0.000

Effect size: rbc -1.00

Plot: next page



Step: Donor2 experiment gel: day1 vs day2

ID: p_d2_exp_day1_vs_day2

Type:

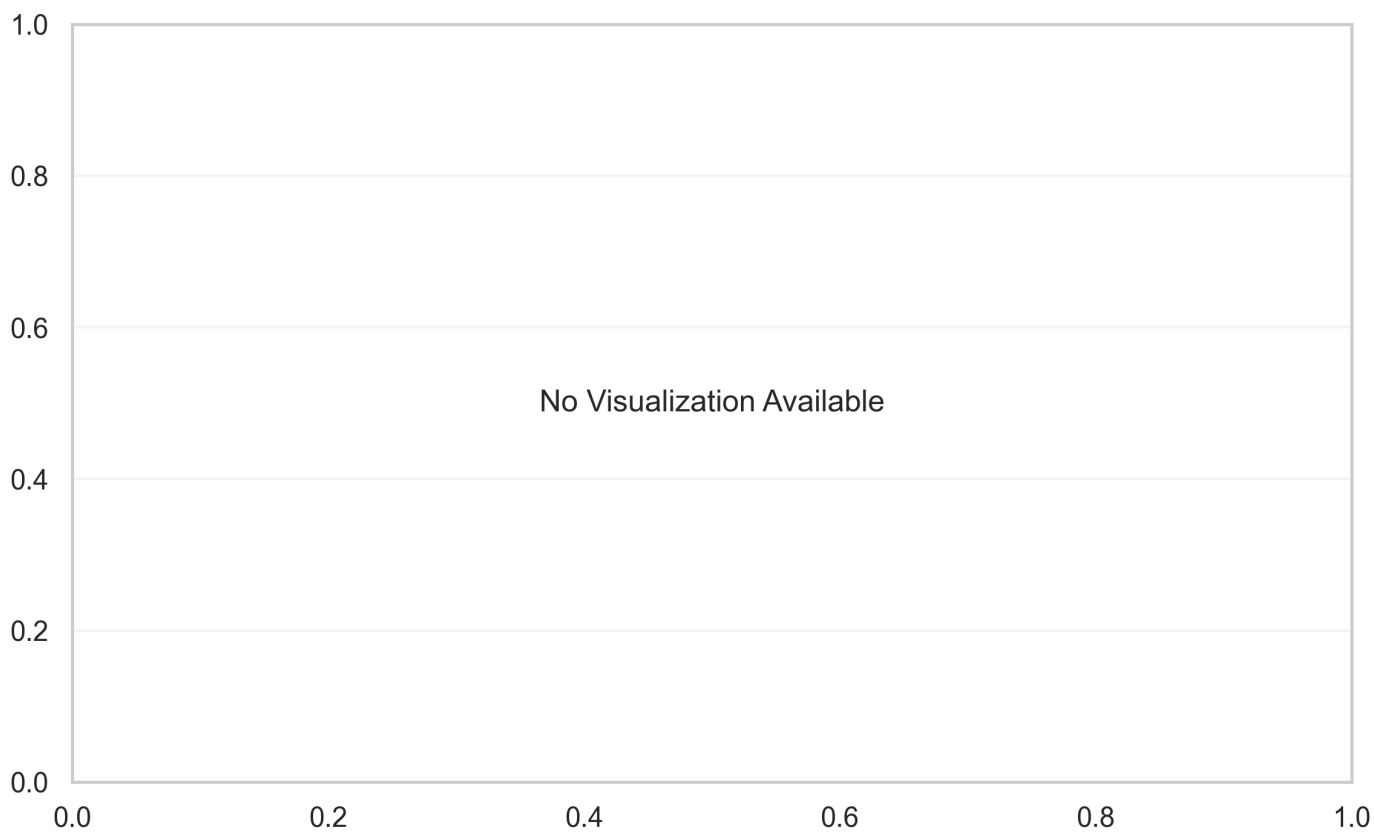
Method: paired wide

p-value: 0.1250

Statistic: 0.000

Effect size: rbc -1.00

Plot: next page



Step: Bland-Altman donor1 gel day1

ID: ba_d1_gel_d1

Type:

Method: Bland-Altman analysis

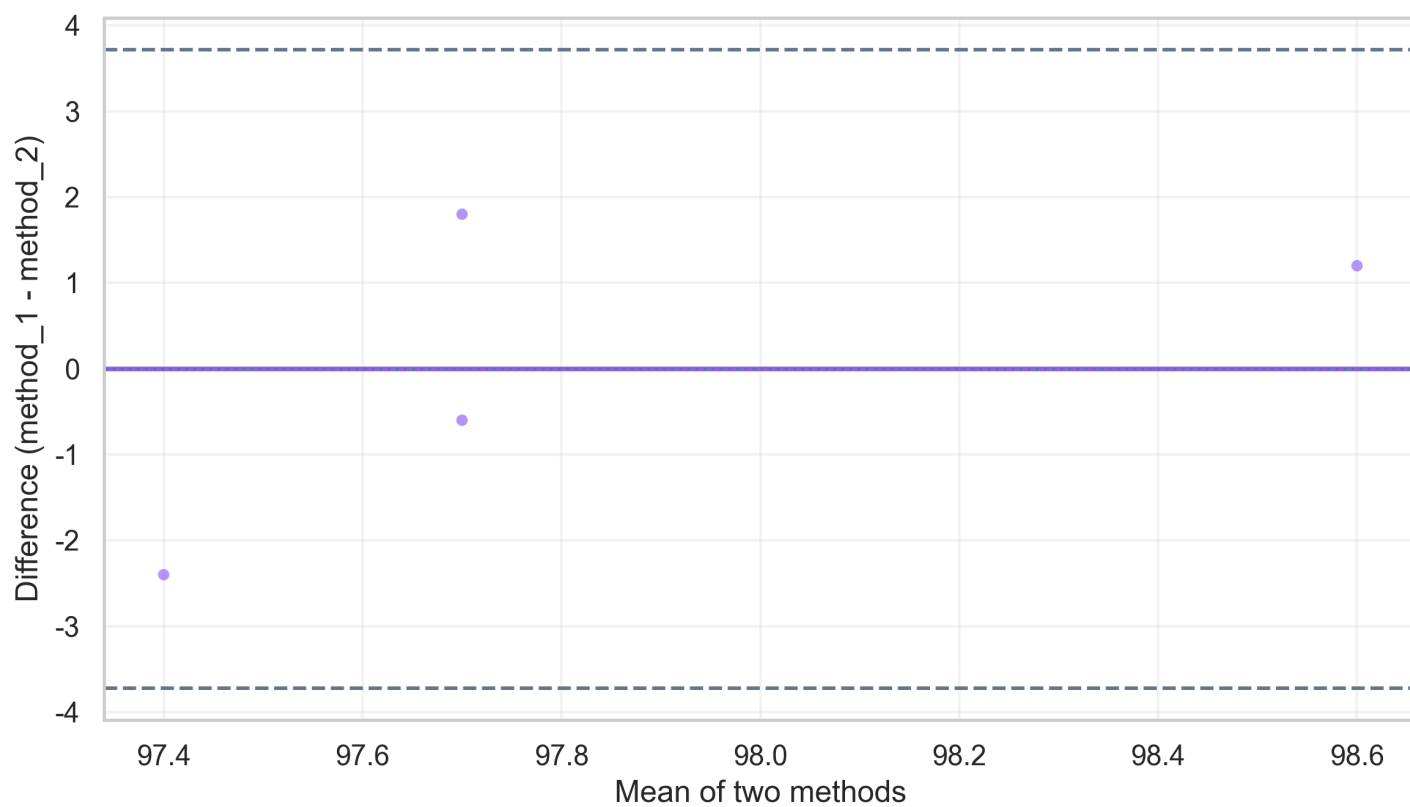
p-value: 1.0000

Statistic: -0.000

Plot: next page

Conclusion: Смещение между методами: -0.000 (95% ДИ [-3.019; 3.019]). Границы согласия: [-3.719; 3.719], вне границ 0/4 (0.0%). Пропорциональное смещение не выявлено. Итоговая оценка согласия: хорошее.

Bland-Altman Plot



Step: Bland-Altman donor2 gel day1

ID: ba_d2_gel_d1

Type:

Method: Bland-Altman analysis

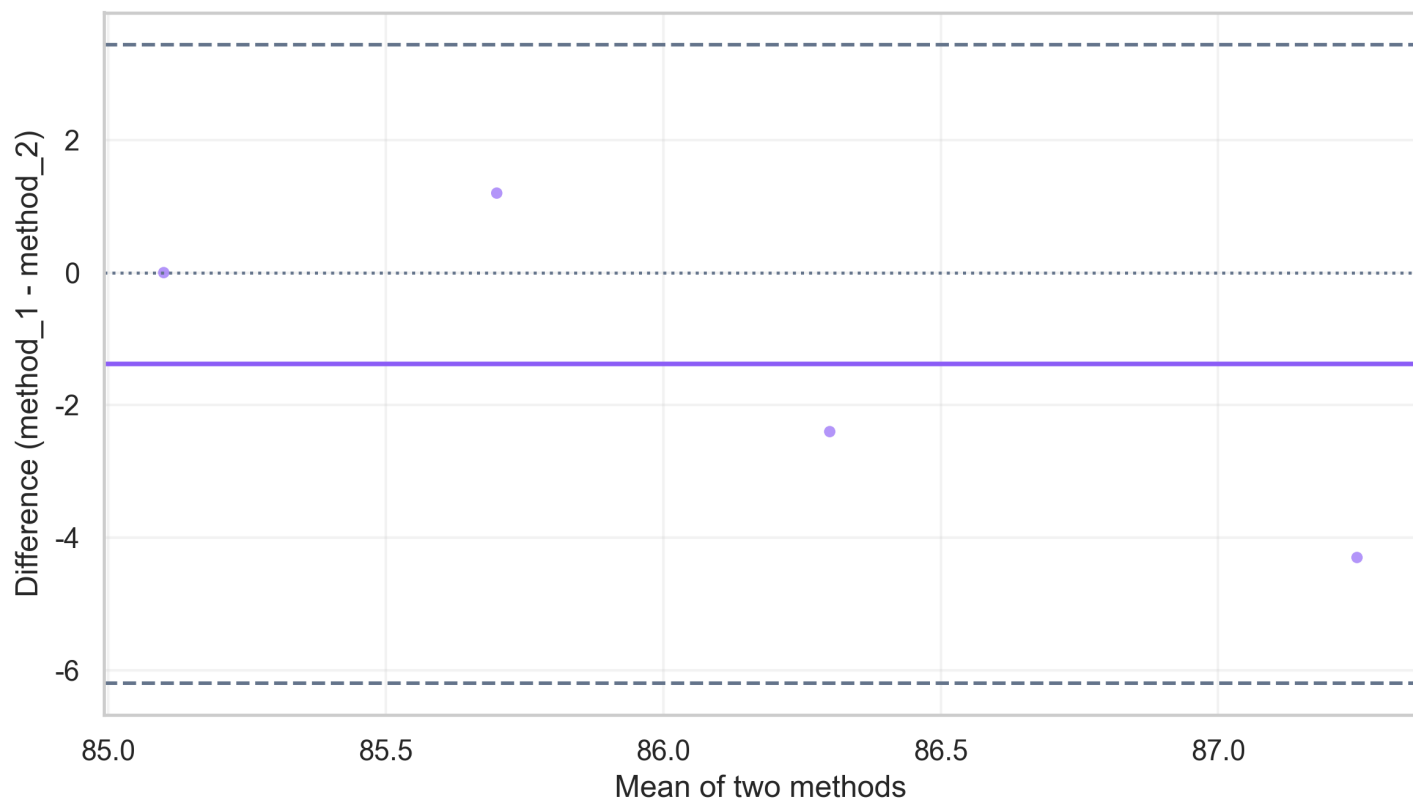
p-value: 0.3448

Statistic: -1.119

Plot: next page

Conclusion: Смещение между методами: -1.375 (95% ДИ [-5.286; 2.536]). Границы согласия: [-6.193; 3.443], вне границ 0/4 (0.0%). Пропорциональное смещение не выявлено. Итоговая оценка согласия: хорошее.

Bland-Altman Plot



Step: Bland-Altman donor1 gel day2

ID: ba_d1_gel_d2

Type:

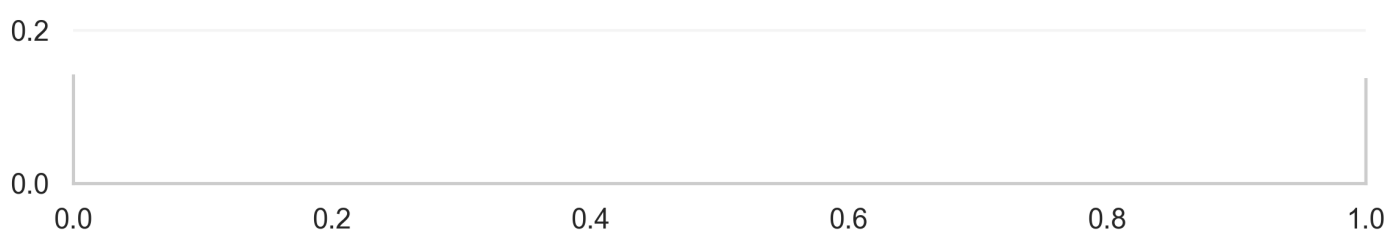
Method: Bland-Altman analysis

p-value: 0.0989

Statistic: -2.365

Plot: next page

Conclusion: Смещение между методами: -1.525 (95% ДИ [-3.577; 0.527]). Границы согласия: [-4.052; 1.002], вне границ 0/4 (0.0%). Пропорциональное смещение не выявлено. Итоговая оценка согласия: хорошее.



Step: Bland-Altman donor2 gel day2

ID: ba_d2_gel_d2

Type:

Method: Bland-Altman analysis

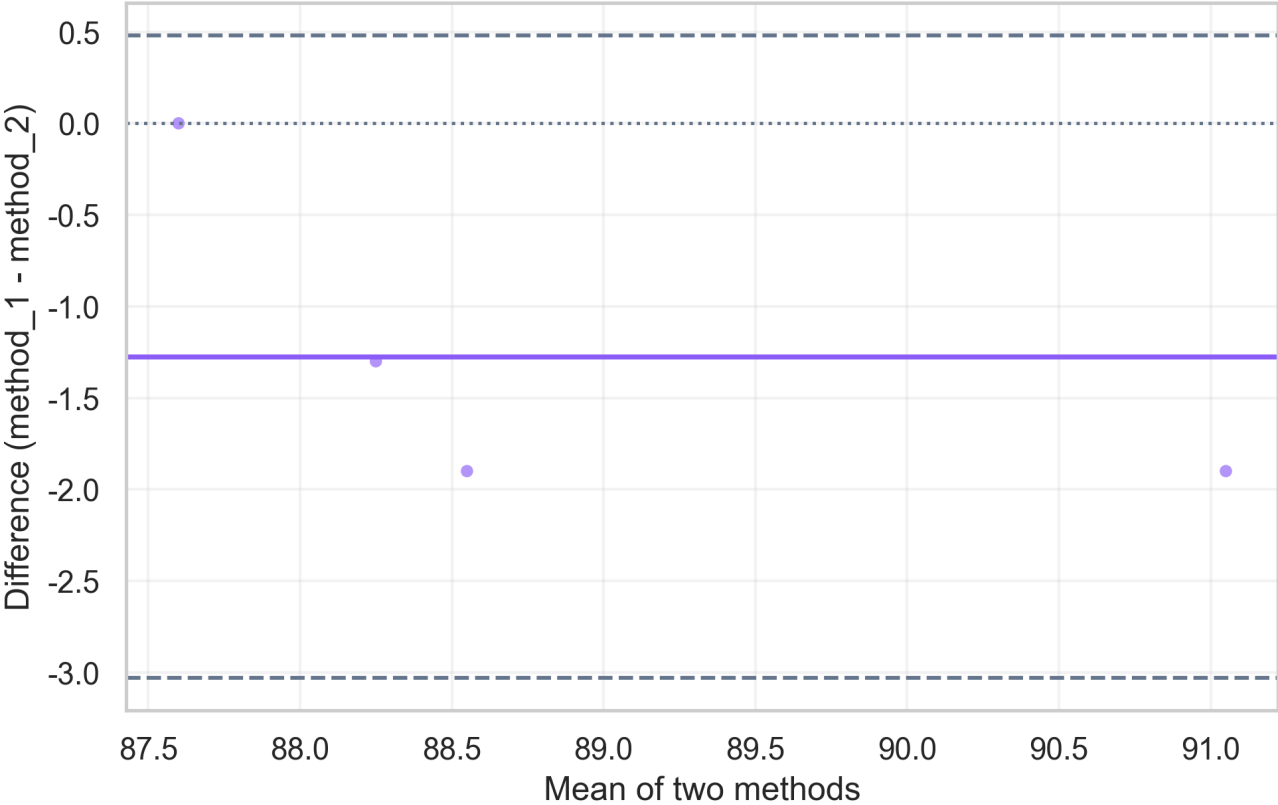
p-value: 0.0653

Statistic: -2.847

Plot: next page

Conclusion: Смещение между методами: -1.275 (95% ДИ [-2.700; 0.150]). Границы согласия: [-3.031; 0.481], вне границ 0/4 (0.0%). Пропорциональное смещение не выявлено. Итоговая оценка согласия: хорошее.

Bland-Altman Plot



Discussion

This report summarizes the findings from 14 statistical analyses, with a focus on agreement between methods and hypothesis testing. All analyses were conducted using an alpha level of 0.05.

No statistically significant results were observed across any of the performed tests, indicating a lack of strong evidence for the investigated effects or differences.

Specifically, Bland-Altman analyses for method agreement consistently showed good agreement, with no proportional bias detected and all observations falling within the limits of agreement.

Hypothesis tests, including paired comparisons, also did not yield significant p-values, suggesting no statistically discernible differences or changes under the tested conditions.

Conclusions

- No statistically significant findings were identified in this analysis.
- Method agreement was consistently rated as good.
- No proportional bias was detected in method comparisons.
- All observations in agreement analyses fell within the limits of agreement.
- Hypothesis tests did not reveal significant differences.

Limitations

- Effect sizes are not available for all inferential steps.
- No statistically significant effects were detected at the current alpha level.